MLOPS ASSIGNMENT-1

Apache Pig

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PROBLEM STATEMENT

A university has a dataset of student performance records containing student ID, department, and marks. Some records contain invalid negative marks or missing marks. The objective is to use Apache Pig and Hadoop to filter these invalid records, group the valid students by department, and calculate the average marks for each department.

DATASET

```
S001,CSE,85
S002,ECE,78
S003,CSE,92
S004,ISE,88
S005,ECE,75
S006,CSE,90
S007,ISE,95
S008,ECE,82
S009,CSE,-5
S010,ISE,
```

IMPLEMENTING CODE

```
students = LOAD '/user/aiml/students/students.csv'
USING PigStorage(',')
AS (student_id:chararray, department:chararray, marks:float);

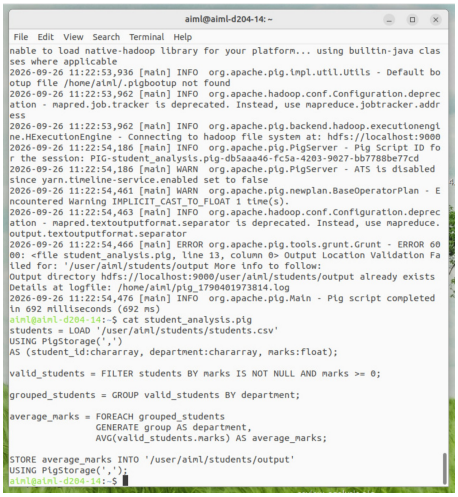
valid_students = FILTER students BY marks IS NOT NULL AND marks >= 0;

grouped_students = GROUP valid_students BY department;

average_marks = FOREACH grouped_students
  GENERATE group AS department,
  AVG(valid_students.marks) AS average_marks;

STORE average_marks INTO '/user/aiml/students/output'
USING PigStorage(',');
```

TERMINAL IMPLEMENTATION



STEPS

1. Create the student dataset with student ID, department, and marks.

2. Store the dataset in HDFS using Hadoop commands.

3. Load the data in Pig using LOAD and PigStorage(',').

4. Filter invalid records where marks are negative or null.

5. Group valid student records based on department.

6. Calculate average marks for each department using AVG().

7. Run the Pig script using Hadoop MapReduce through YARN.

8. Store the resulting department-wise averages in HDFS and copy the result to a text file.

TOOLS

Tool / Technology	Purpose
Ubuntu Linux	Operating system / terminal
Java 8	Runtime for Hadoop and Pig
Apache Hadoop 3.3.6	Distributed data processing framework
HDFS	Stores input dataset and output
YARN / MapReduce	Executes the Pig job
Apache Pig 0.17.0	Data-flow processing using Pig Latin
Pig Latin	Scripting language used for analysis

WORKFLOW

students.csv → **HDFS** → **LOAD** → **FILTER** → **GROUP BY department** → **AVG** → **STORE** → **student_output.txt**

RESULT

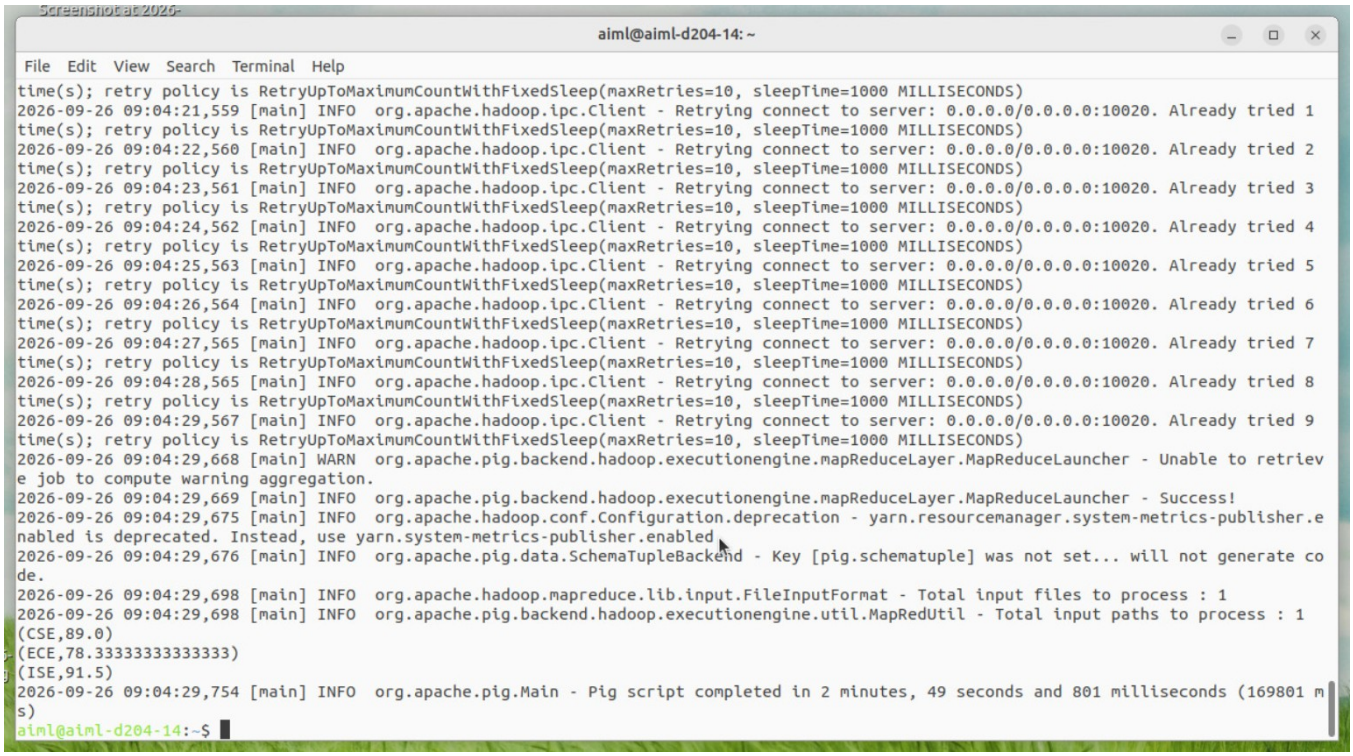
The Apache Pig program successfully filtered invalid student marks and calculated the average marks for each department. The valid records were grouped by department and the results were stored in HDFS and copied to a text file. The final averages were CSE = 89.0, ECE = 78.33, and ISE = 91.5.

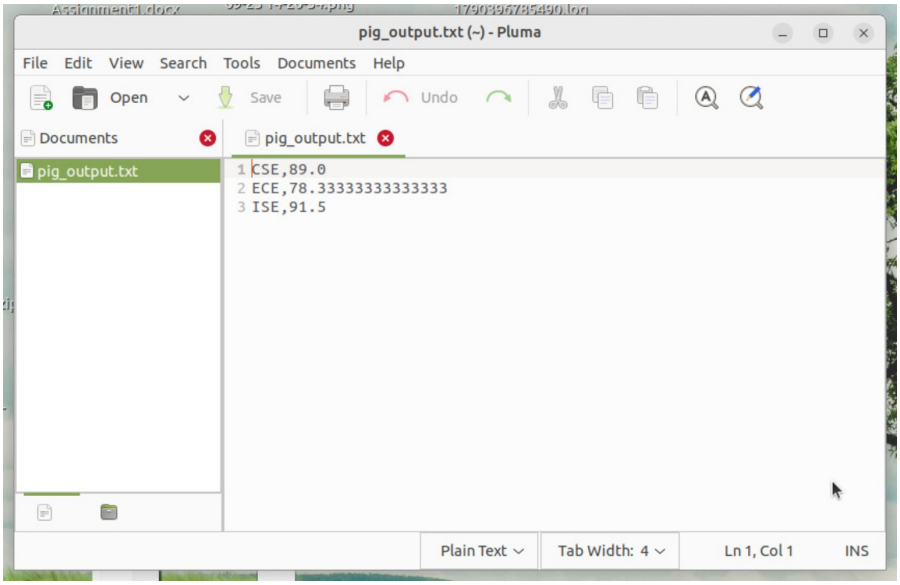
CONCLUSION

The Apache Pig implementation achieved the objective of the student performance analysis problem. Pig Latin provided a data-flow approach for loading, filtering, grouping, aggregating, and storing the data, while Hadoop HDFS and YARN supported storage and execution.

OUTPUT

The Pig program was executed using Hadoop MapReduce through YARN. The successful execution output and the generated text file are shown below.





FINAL OUTPUT

Department	Average Marks
CSE	89.0
ECE	78.33333333333333
ISE	91.5